

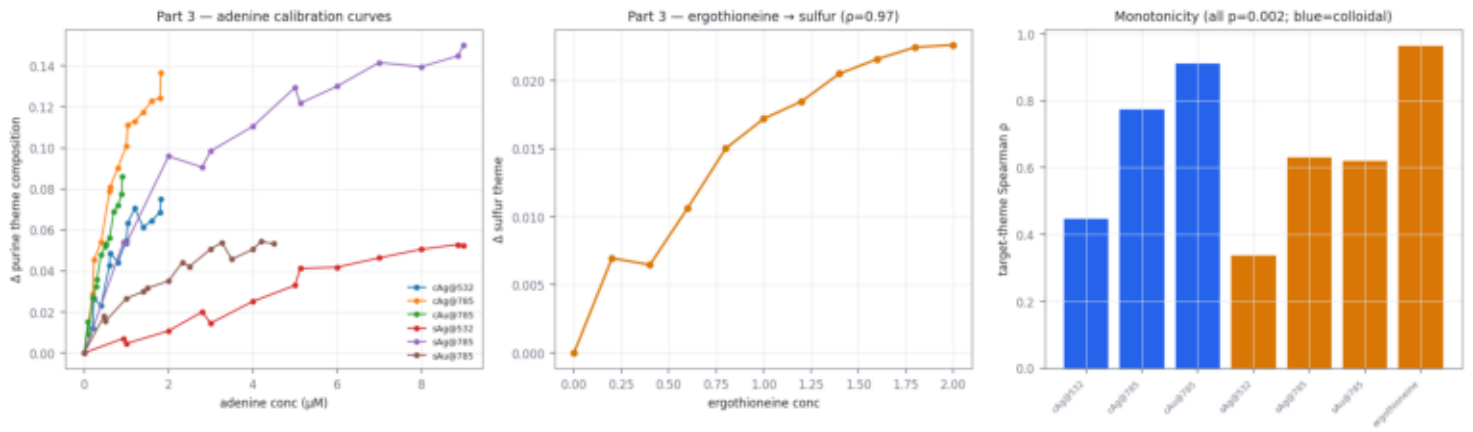
Biochemical State Vector — Validation

GAIRA V6 — characterizing the BSV as a scientific coordinate system

*The V6 engine, atlas, ontology and weights are FROZEN and only MEASURED here.
The question is not whether the software runs, but whether the BSV is a stable,
meaningful biochemical coordinate system.*

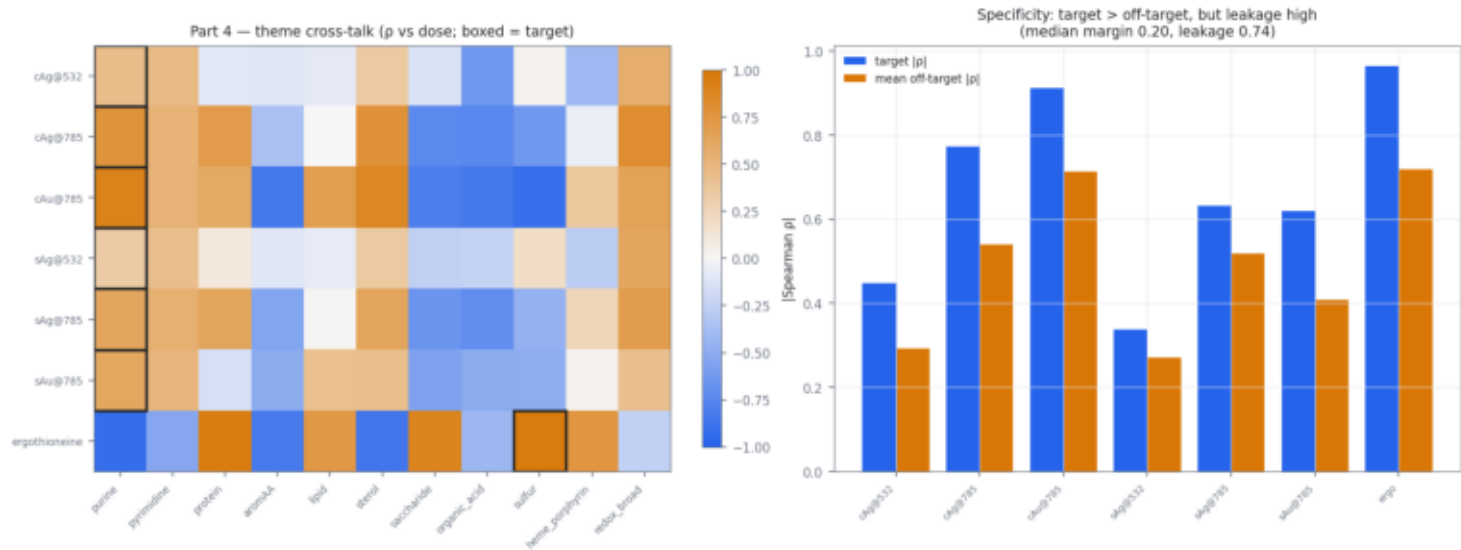
Monotonicity (dose→target theme)	monotonic + saturating, all $p=0.002$; ρ 0.34–0.97
Best calibration	ergothioneine→sulfur ρ 0.97; colloidal adenine→purine ρ 0.91
Theme specificity	target > off-target, but leakage 0.65–0.82 (high cross-talk)
Theme coupling	mean $ r $ 0.24, max 0.65 — and it ENCODES biology
Effective dimensionality	~4 of 11 themes (PC1 = 50% variance)
Inter-analyte geometry	purines cluster; glucose isomers merge
Confidence	tracks domain OOD (ρ –0.57); NOT analyte recoverability
Reproducibility	substrate-dependent (ICC 0.14 paper → 0.83 colloid)

Calibration



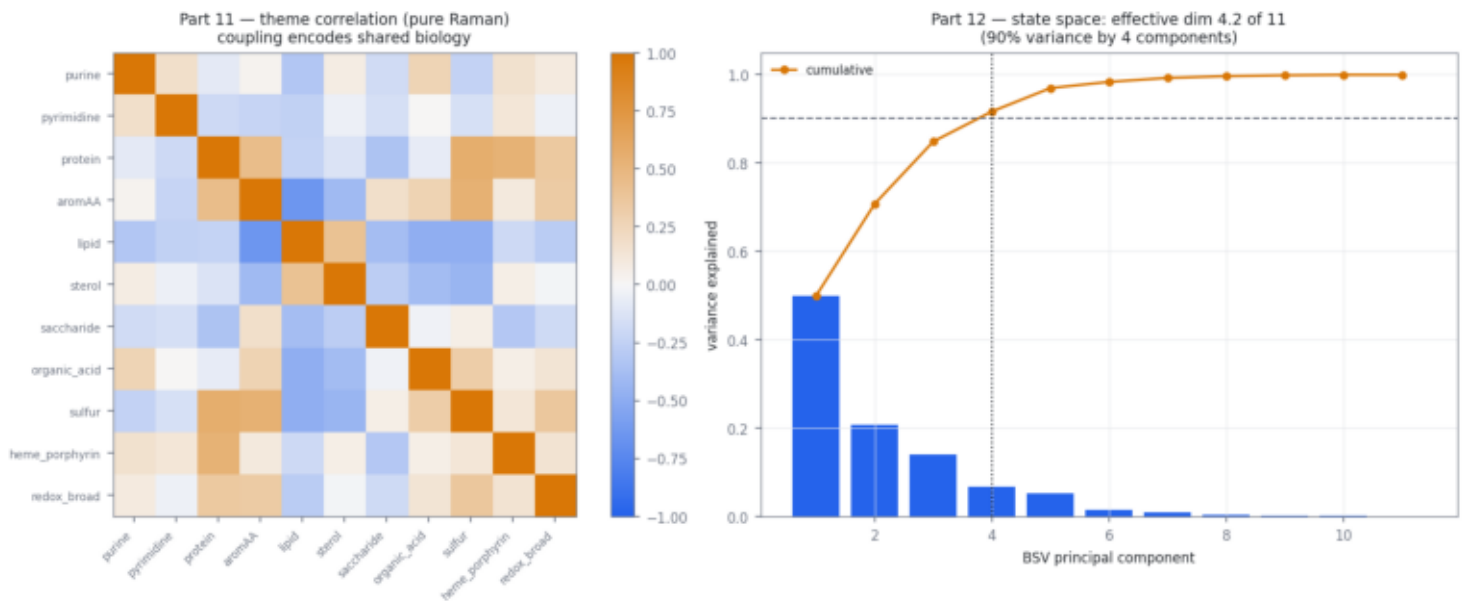
Calibration — dose→theme curves are monotonic and saturating.

Theme cross-talk



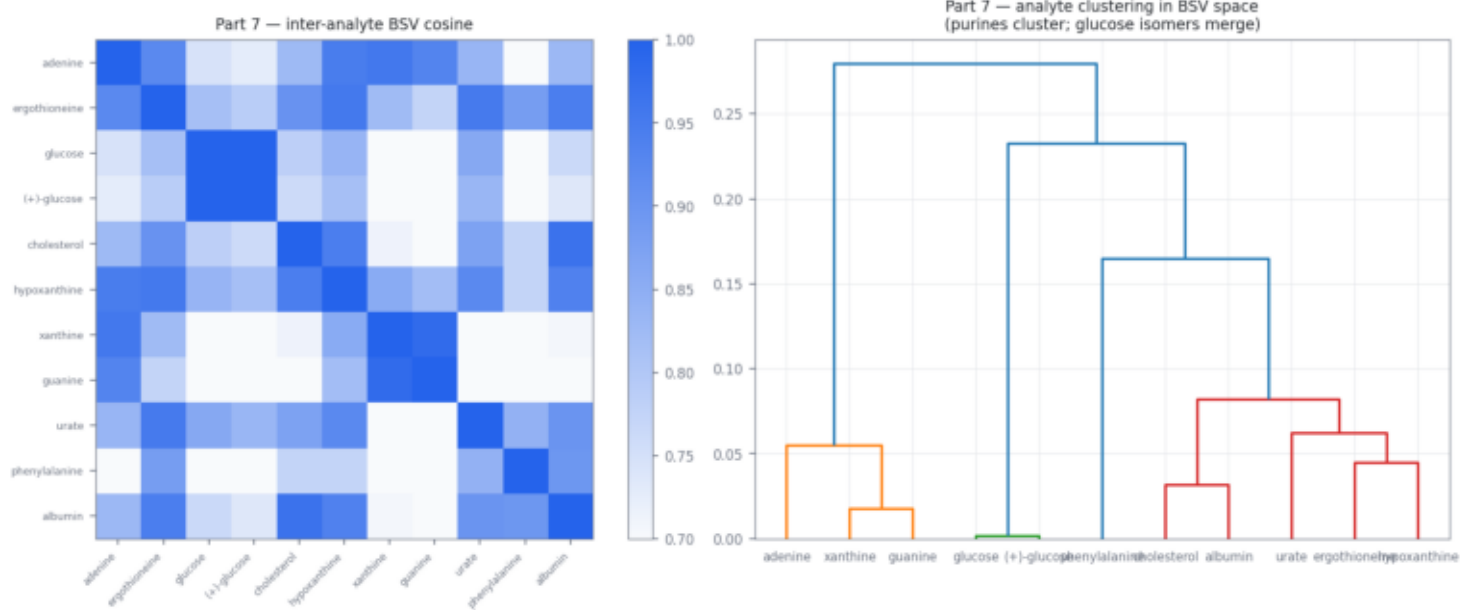
Theme cross-talk — target moves most, but off-target leakage is high.

Theme coupling encodes biology; the space is effectively ~4-D.



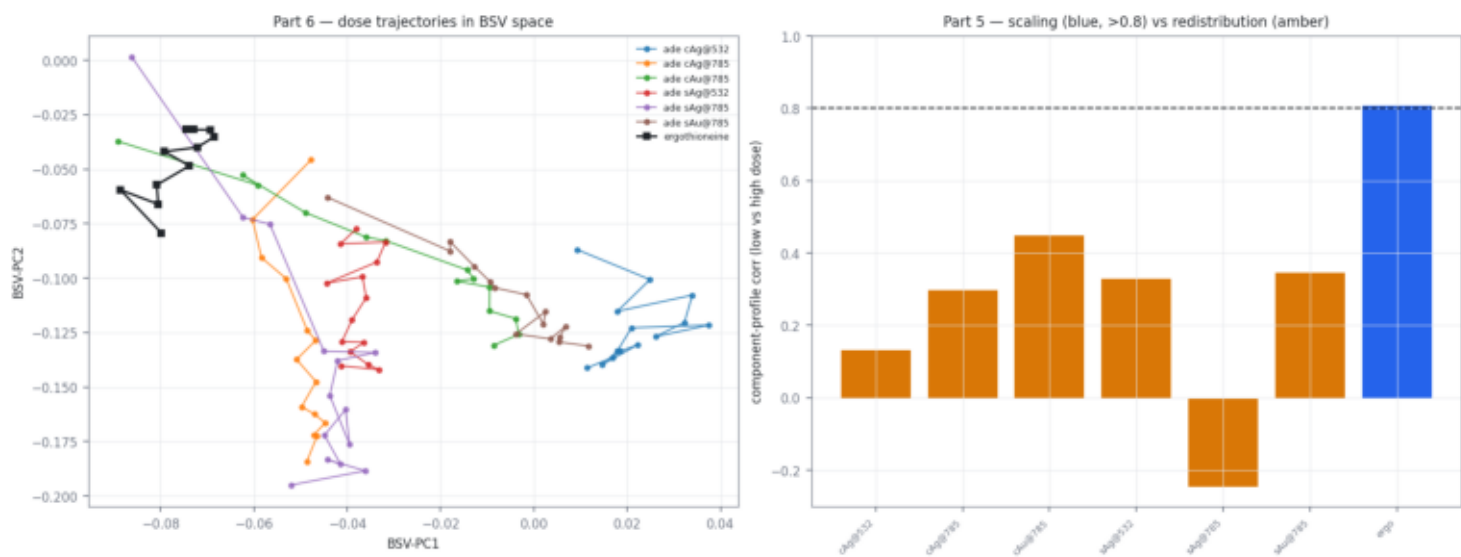
Theme coupling encodes biology; the space is effectively ~4-D.

Inter-analyte geometry



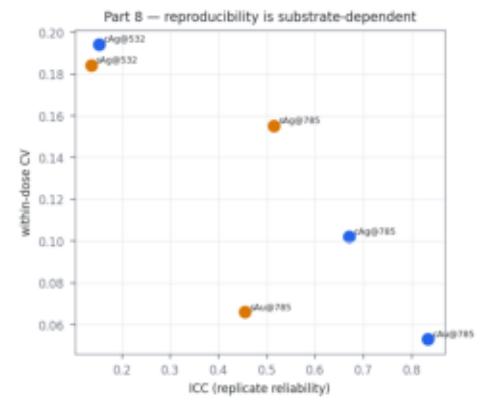
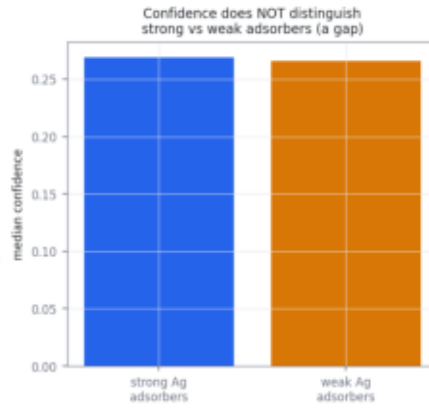
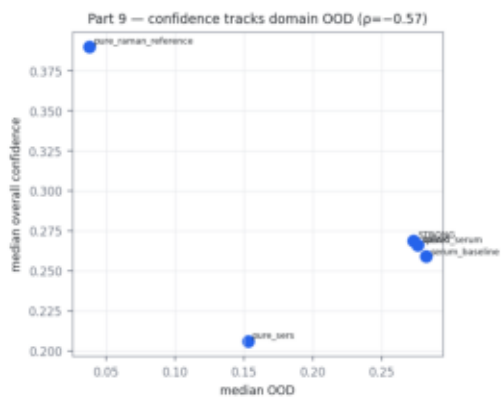
Inter-analyte geometry — purines cluster in BSV space.

Dose trajectories; adenine redistributes, ergothioneine scales.



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Confidence tracks domain OOD but not analyte recoverability; reproducibility



Confidence tracks domain OOD but not analyte recoverability; reproducibility is substrate-dependent.